Vector Calculus (MA1130)

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Preface

These notes have been prepared for the course **Vector Calculus (MA1130)**, a 1-credit course offered during Segments $5-6$ of the Fall Semester 2026 at the Indian Institute of Technology Hyderabad (IITH). They are based on the lectures delivered in the course, and their primary purpose is to provide a concise and organized presentation of the topics prescribed in the course syllabus.

These notes are neither self-contained nor intended to serve as a comprehensive textbook on calculus. There are many excellent textbooks on the subject that offer a more comprehensive and detailed treatment. Students are encouraged to consult these standard references for additional examples, exercises, and deeper insights.

Many of the definitions, examples, exercises, and proofs presented in these notes have been adapted from or inspired by several standard texts, including *Vector Analysis and an Introduction to Tensor Analysis* by Murray R. Spiegel and lecture notes named *Vector Calculus* by Michael Corral, Prof. Sukumar D, Prof. M Rajesh Kannan, among others. I gratefully acknowledge these valuable sources.

The material presented here is intended to complement the classroom lectures by emphasizing the fundamental concepts, definitions, theorems, proofs, and illustrative examples discussed during the course. Every effort has been made to keep the exposition clear, concise, and accessible to first-year B.Tech students.

Despite every effort to ensure accuracy, these notes may still contain typographical errors or other inadvertent mistakes. I would be grateful if readers could bring any such errors or suggestions for improvement to my attention by email.

Finally, I would like to acknowledge the assistance of AI tools in preparing the LaTeX typesetting and formatting of these notes, which significantly reduced the time required for their preparation.

Introduction

Calculus is one of the most important tools in mathematics. In elementary calculus, we study functions of a single variable such as $f(x)$, where the value of the function depends on only one quantity. Many real-world phenomena depend on several variables. For example, the temperature in a room depends on the position of a point and may be represented by $T(x, y, z)$, while the height of a landscape can be described by a function $z = f(x, y)$. Such functions are studied in multivariable calculus.

In many applications, it is not enough to describe only the magnitude of a quantity. We often need both its magnitude and its direction. Quantities of this type are called *vectors*. Examples include velocity, force, acceleration, and electric field. A collection of vectors assigned to every point of a region is called a *vector field*. For instance, the velocity of water in a river or the wind velocity in the atmosphere can be represented by vector fields.

One of the fundamental questions in multivariable calculus is to understand how scalar and vector fields change from one point to another. For a scalar field such as the temperature $T(x, y, z)$, we may ask:

> In which direction does the temperature increase most rapidly?

The mathematical tool that answers this question is the *gradient*. Similarly, for a vector field, we are interested in measuring how the field spreads out from a point or how it rotates around a point. These ideas lead to the notions of *divergence* and *curl*.

Integration also plays an important role in several variables. Double and triple integrals are used to compute areas, volumes, masses, and other physical quantities. In vector calculus, integration is extended further. Instead of integrating over intervals or regions, we may integrate along curves or over surfaces. Such integrals arise naturally in applications. For example, the work done by a force $\mathbf{F}$ in moving an object along a curve C is given by $\int_C \mathbf{F} \cdot d\mathbf{r}$, while the flow of a fluid through a surface S is measured by the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$.

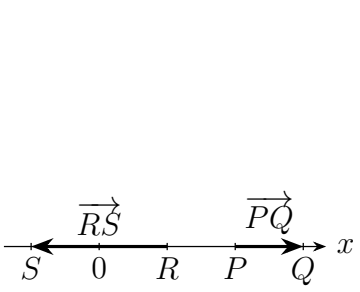
A remarkable feature of vector calculus is that differentiation and integration are closely related. Some of the most important results in mathematics establish deep connections between line integrals, surface integrals, and multiple integrals. These results form the foundation of many applications in mathematics, physics, and engineering.

This course aims to introduce the basic concepts of multivariable and vector calculus, develop techniques for computing multiple and vector integrals, and study the fundamental ideas that connect them. Throughout the course, mathematical concepts will be motivated and illustrated through a variety of examples.

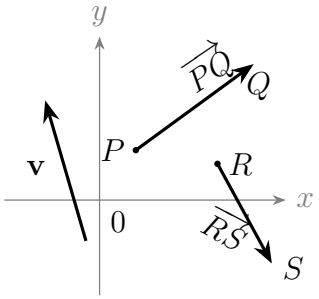
1 Vectors in Euclidean space

1.1 Vector algebra

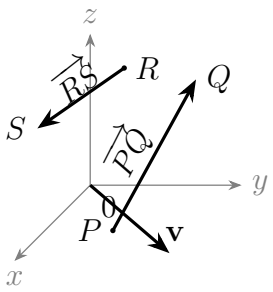
A ***vector*** is a quantity having both magnitude and direction, such as displacement, velocity, force, and acceleration. Graphically, a vector is represented by a directed line segment (or an arrow) $\overrightarrow{PQ}$. The direction of the arrow specifies the direction of the vector, while its length represents the magnitude of the vector. The tail end P of the arrow is called the *origin* or *initial point* of the vector, and the head Q is called the *terminal point* or *terminus*.



(a) One dimension



(b) Two dimensions



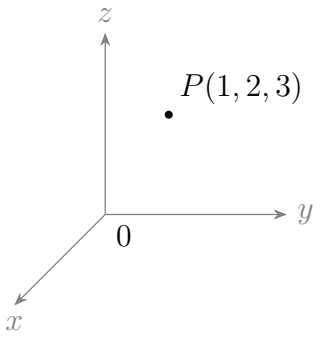
(c) Three dimensions

Definition 1. *A* (nonzero) vector *is a directed line segment from a point P, called the* initial point*, to a distinct point Q, called the* terminal point*. The vector is denoted by $\overrightarrow{PQ}$. Its* magnitude *is the length of the directed line segment and is denoted by $\|\overrightarrow{PQ}\|$. The* direction *of the vector is the direction from P to Q. The* zero vector *has magnitude zero and is denoted by $\mathbf{0}$.*

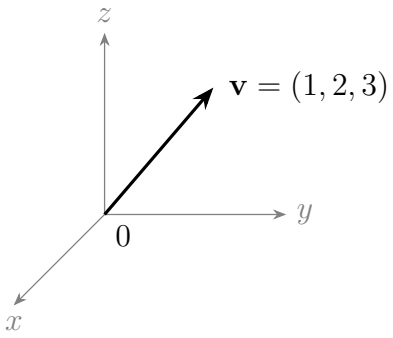
- The zero vector has magnitude 0, that is, $\|\mathbf{0}\| = 0$. Since it has no well-defined direction, the notion of direction is defined only for nonzero vectors.

Definition 2. *Two nonzero vectors are said to be* equal *if they have the same magnitude and the same direction. Any vector with zero magnitude is equal to the zero vector.*

- Unless stated otherwise, a vector will be assumed to have its initial point at the origin of the coordinate system.



(a) The point $(1, 2, 3)$.



(b) The vector $(1, 2, 3)$.

Let $P = (x_1, y_1, z_1)$ and $Q = (x_2, y_2, z_2)$ be two points in $\mathbb{R}^3$. The vector from P to Q is defined by

$$\overrightarrow{PQ} = Q - P = (x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1).$$

The magnitude of $\overrightarrow{PQ}$ is

$$\|\overrightarrow{PQ}\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2},$$

which is precisely the distance between the points P and Q. Hence,

$$d(P, Q) = \|\overrightarrow{PQ}\|.$$

A ***scalar*** is a quantity having magnitude but no direction, such as mass, length, time, temperature, and any real number. Scalars are denoted by ordinary letters, and operations involving scalars follow the usual rules of elementary algebra.

Definition 3. *Let k be a scalar and $\mathbf{v}$ be a nonzero vector. The* scalar multiple *of $\mathbf{v}$ by k, denoted by $k\mathbf{v}$, is the vector whose magnitude is $|k|\,\|\mathbf{v}\|$. Moreover,*

- *if $k > 0$, then $k\mathbf{v}$ has the same direction as $\mathbf{v}$;*
- *if $k < 0$, then $k\mathbf{v}$ has the opposite direction to $\mathbf{v}$;*
- *if $k = 0$, then $k\mathbf{v} = \mathbf{0}$.*

For the zero vector, we define $k\mathbf{0} = \mathbf{0}$ for every scalar k.

- The algebra of vectors, including vector addition and subtraction, is assumed to be familiar from the plus two level coordinate geometry and vectors. Readers are encouraged to briefly review these topics before proceeding.

Definition 4. *A vector* $\mathbf{u}$ *is called a* unit vector *if its magnitude is* 1*, that is,*

$$\|\mathbf{u}\| = 1.$$

If $\mathbf{v}$ *is a nonzero vector, then the unit vector in the direction of* $\mathbf{v}$ *is given by*

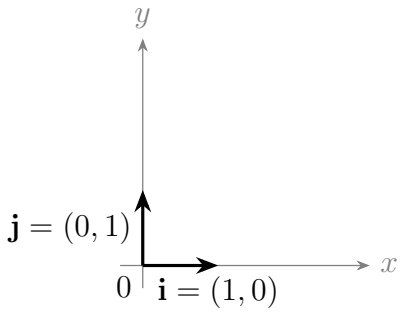
$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

- In $\mathbb{R}^2$, the standard basis vectors are

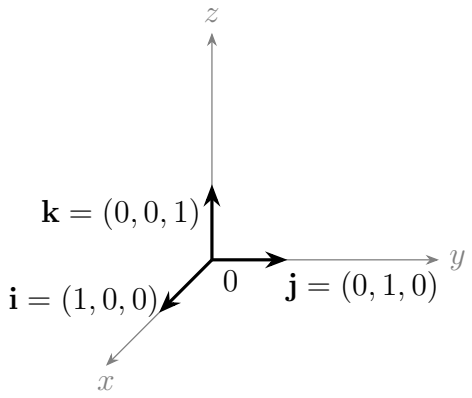
$$\mathbf{i} = (1, 0), \qquad \mathbf{j} = (0, 1),$$

and in $\mathbb{R}^3$, the standard basis vectors are

$$\mathbf{i} = (1, 0, 0), \qquad \mathbf{j} = (0, 1, 0), \qquad \mathbf{k} = (0, 0, 1).$$



(a) Standard basis of $\mathbb{R}^2$.



(b) Standard basis of $\mathbb{R}^3$.

Let $\mathbf{v}$ be a vector in $\mathbb{R}^3$. Then $\mathbf{v}$ can be expressed uniquely as

$$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k},$$

where v_1, v_2, and v_3 are called the *components* of $\mathbf{v}$ along the x, y, and z axes, respectively. Equivalently,

$$\mathbf{v} = (v_1, v_2, v_3).$$

The magnitude of $\mathbf{v}$ is given by

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

The *position vector* (or *radius vector*) of a point $P = (x, y, z)$ is the vector from the origin O to P, denoted by

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = (x, y, z).$$

Its magnitude is

$$\|\mathbf{r}\| = \sqrt{x^2 + y^2 + z^2}.$$

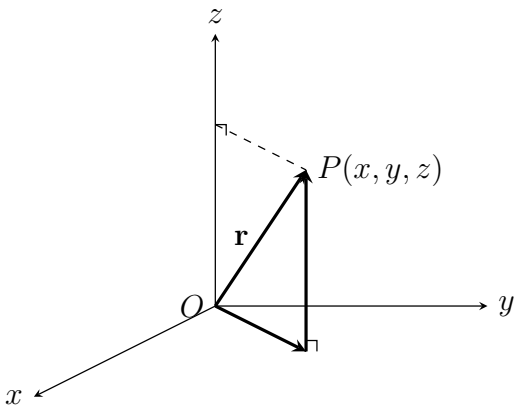
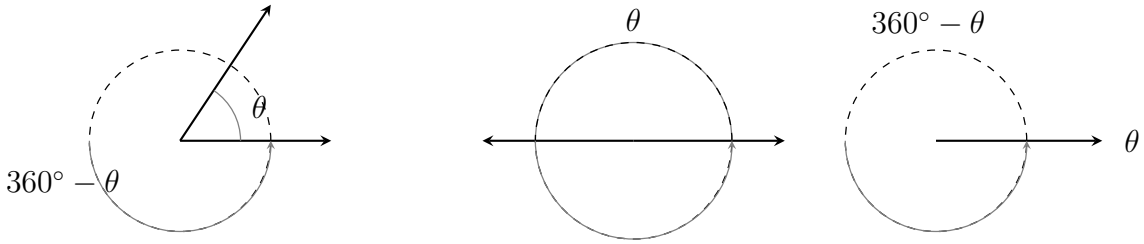


Figure 4: The position vector $\mathbf{r} = \overrightarrow{OP}$ and its components.

1.2 The dot and cross product

The **angle** between two nonzero vectors with the same initial point is the smallest angle between them.



(a) $0° < \theta < 180°$ **(b)** $\theta = 180°$ **(c)** $\theta = 0°$

Figure 5: The angle between two vectors.

The angle between the zero vector and any vector is not defined. For two nonzero vectors with the same initial point, there are two possible angles between them, namely θ and $360° - \theta$. Unless stated otherwise, the *angle between two vectors* will always mean the smaller angle, where

$$0° \leq \theta \leq 180°.$$

Definition 5. *Let* $\mathbf{v} = (v_1, v_2, v_3)$ *and* $\mathbf{w} = (w_1, w_2, w_3)$ *be vectors in* $\mathbb{R}^3$*. The* dot product *(or* scalar product*) of* $\mathbf{v}$ *and* $\mathbf{w}$*, denoted by* $\mathbf{v} \cdot \mathbf{w}$*, is defined by*

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3.$$

Theorem 1. *Let* $\mathbf{v}$ *and* $\mathbf{w}$ *be nonzero vectors, and let* $\theta \in [0, \pi]$ *denote the angle between them. Then*

$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\| \, \|\mathbf{w}\| \cos\theta.$$

In particular, $\mathbf{v}$ *and* $\mathbf{w}$ *are perpendicular if and only if*

$$\mathbf{v} \cdot \mathbf{w} = 0.$$

- Readers are encouraged to briefly review the basic properties of the dot product from the plus two level before proceeding.

Definition 6. *Let* $\mathbf{v} = (v_1, v_2, v_3)$ *and* $\mathbf{w} = (w_1, w_2, w_3)$ *be vectors in* $\mathbb{R}^3$*. The* cross product *(or* vector product*) of* $\mathbf{v}$ *and* $\mathbf{w}$*, denoted by* $\mathbf{v} \times \mathbf{w}$*, is defined by*

$$\mathbf{v} \times \mathbf{w} = (v_2 w_3 - v_3 w_2,\ v_3 w_1 - v_1 w_3,\ v_1 w_2 - v_2 w_1).$$

Theorem 2. *Let* $\mathbf{v}$ *and* $\mathbf{w}$ *be nonzero vectors. If* $\mathbf{v} \times \mathbf{w} \neq \mathbf{0}$*, then* $\mathbf{v} \times \mathbf{w}$ *is perpendicular to both* $\mathbf{v}$ *and* $\mathbf{w}$*.*

Theorem 3. *Let* $\mathbf{v}$ *and* $\mathbf{w}$ *be nonzero vectors in* $\mathbb{R}^3$*, and let* θ *denote the angle between them. Then*

$$\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \, \|\mathbf{w}\| \sin\theta.$$

- Readers are encouraged to review the basic properties of the cross product, its relationship with the dot product, and the standard geometric applications of vector algebra, including formulas for the area of a triangle, the area of a parallelogram, and the volume of a parallelepiped, from the plus two level before proceeding.

1.3 Lines, planes and surfaces

In this section, we study how vectors can be used to represent lines in three-dimensional Euclidean space.

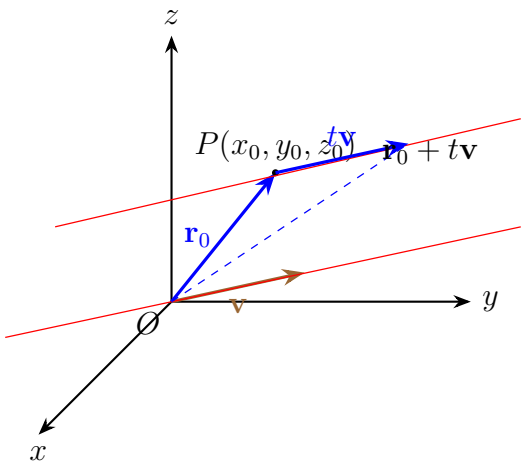


Figure 6: Vector equation of a line.

Let $P = (x_0, y_0, z_0)$ be a point in $\mathbb{R}^3$, and let

$$\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} = (a, b, c) \neq \mathbf{0}$$

be a direction vector. Then the line passing through P and parallel to $\mathbf{v}$ can be represented in the following equivalent forms:

(i) **Vector equation:**

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}, \qquad t \in \mathbb{R},$$

where $\mathbf{r}_0 = x_0\mathbf{i} + y_0\mathbf{j} + z_0\mathbf{k}$ is the position vector of P.

(ii) **Parametric equations:**

$$x = x_0 + at, \qquad y = y_0 + bt, \qquad z = z_0 + ct, \qquad t \in \mathbb{R}.$$

(iii) **Symmetric equations:** If a, b, and c are all nonzero, then

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

- A *parametric representation* of a curve is a representation in which the coordinates of every point on the curve are expressed as functions of a parameter.

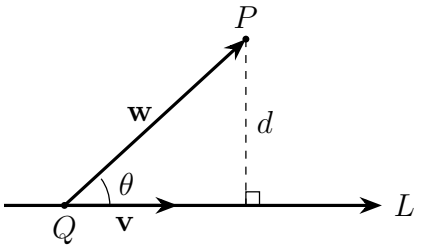


Figure 7: Distance from a point to a line.

Let

$$L: \ \mathbf{r} = \mathbf{r}_0 + t\mathbf{v}, \qquad t \in \mathbb{R},$$

be a line in $\mathbb{R}^3$, where $\mathbf{v} \neq \mathbf{0}$, and let P be a point not lying on L. Choose any point Q on L, and let

$$\mathbf{w} = \overrightarrow{QP}.$$

Then the perpendicular distance from P to L is given by

$$d(P, L) = \frac{\|\mathbf{v} \times \mathbf{w}\|}{\|\mathbf{v}\|}.$$

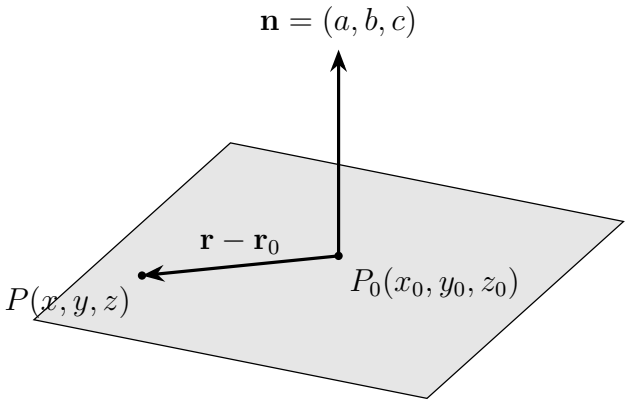


Figure 8: A plane with normal vector $\mathbf{n}$.

Let $P_0 = (x_0, y_0, z_0)$ be a point in $\mathbb{R}^3$, and let

$$\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} = (a, b, c) \neq \mathbf{0}$$

be a normal vector to a plane. Then the plane passing through P_0 with normal vector $\mathbf{n}$ can be represented in the following equivalent forms:

(i) **Vector equation:**

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0,$$

where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is the position vector of an arbitrary point (x, y, z) on the plane, and $\mathbf{r}_0 = x_0\mathbf{i} + y_0\mathbf{j} + z_0\mathbf{k}$ is the position vector of P_0.

(ii) **Point-normal form:**

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

(iii) **Normal form:**

$$ax + by + cz + d = 0.$$

Let $Q = (x_0, y_0, z_0)$ be a point in $\mathbb{R}^3$, and let

$$ax + by + cz + d = 0$$

be the equation of a plane not containing Q. Then the perpendicular distance from Q to the plane is given by

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

In the previous part, we studied lines and planes in Euclidean space. We now turn our attention to *surfaces* in $\mathbb{R}^3$. Informally, a surface is the set of all points (x, y, z) satisfying an equation of the form

$$F(x, y, z) = 0,$$

where $F : \mathbb{R}^3 \to \mathbb{R}$ is a given function. Thus, the plane

$$ax + by + cz + d = 0$$

is a surface corresponding to the function

$$F(x, y, z) = ax + by + cz + d.$$

In this part, we introduce some of the most common surfaces in $\mathbb{R}^3$ and discuss their equations and graphical representations.

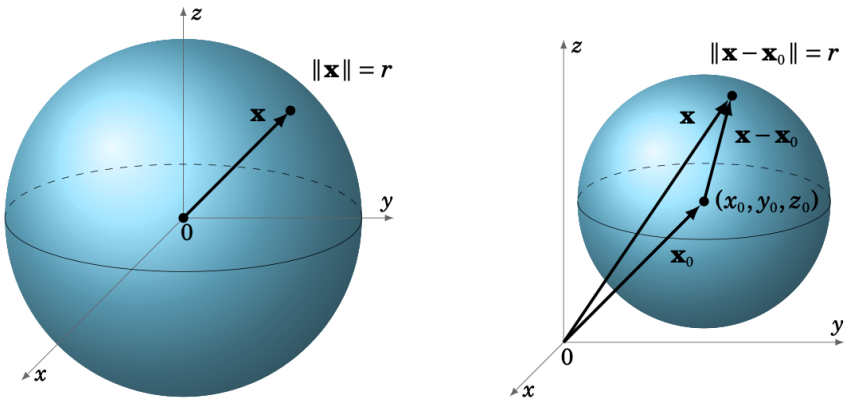
A *sphere* with center $P_0 = (x_0, y_0, z_0)$ and radius $r > 0$ is the set of all points $(x, y, z) \in \mathbb{R}^3$ whose distance from P_0 is equal to r. Thus,

$$S = \left\{(x, y, z) \in \mathbb{R}^3 : (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2\right\}.$$

Equivalently, in vector notation,

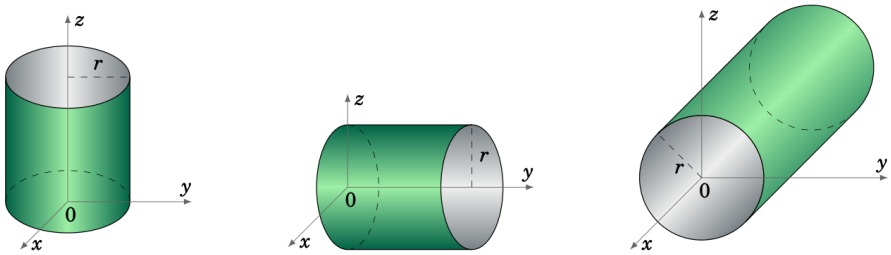
$$S = \left\{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x} - \mathbf{x}_0\| = r\right\},$$

where $\mathbf{x} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $\mathbf{x}_0 = x_0\mathbf{i} + y_0\mathbf{j} + z_0\mathbf{k}$ are the position vectors of (x, y, z) and P_0, respectively.



(a) radius r, center $(0,0,0)$ **(b)** radius r, center (x_0, y_0, z_0)

In this course, we consider only *right circular cylinders*. Such a cylinder is obtained by moving a straight line parallel to itself along a circle in $\mathbb{R}^3$, with the line remaining perpendicular to the plane containing the circle. Throughout this course, we restrict our attention to cylinders whose axes are parallel to one of the coordinate axes.



(a) $x^2 + y^2 = r^2$, any z **(b)** $x^2 + z^2 = r^2$, any y **(c)** $y^2 + z^2 = r^2$, any x

Figure 1.6.3 Cylinders in $\mathbb{R}^3$

An *ellipsoid* is the set

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \right\},$$

where $a, b, c > 0$.

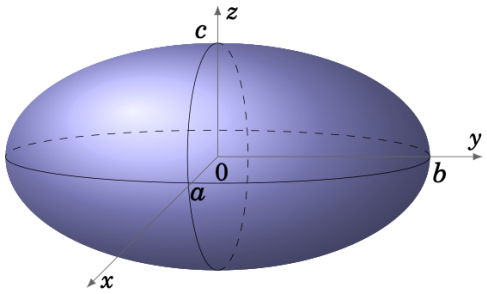


Figure 1.6.4 Ellipsoid

A *hyperboloid of one sheet* is the set

$$H = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \right\},$$

where $a, b, c > 0$.

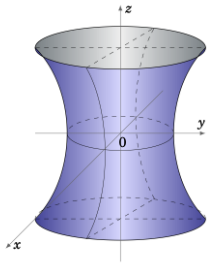


Figure 1.6.5 Hyperboloid of one sheet

A *hyperboloid of two sheets* is the set

$$H = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \right\},$$

where $a, b, c > 0$.

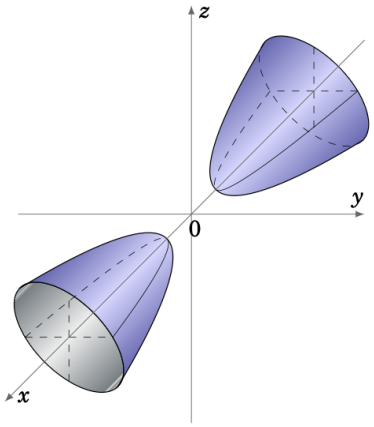


Figure 1.6.6 Hyperboloid of two sheets

An *elliptic paraboloid* is the set

$$P = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c} \right\},$$

where $a, b, c > 0$. If $c < 0$, the paraboloid simply opens downward instead of upward.

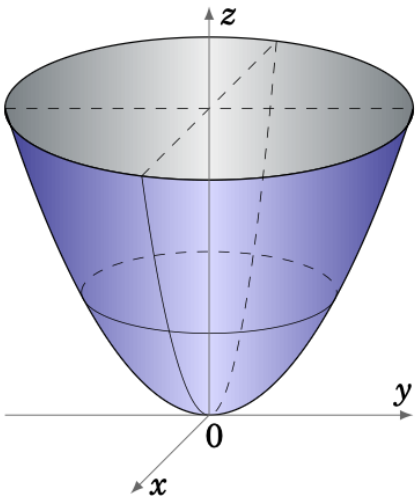


Figure 1.6.7 Paraboloid

An *elliptic cone* is the set

$$C = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0 \right\},$$

where $a, b, c > 0$.

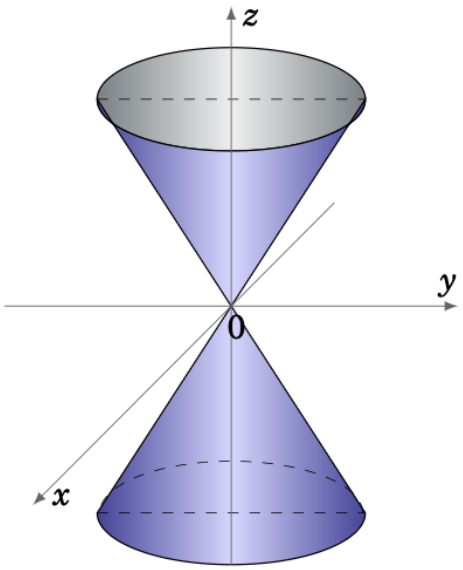


Figure 1.6.9 Elliptic cone

Lecture 2, October 9, 2026

1.4 Curvilinear coordinates

Polar coordinates. In Cartesian coordinates, a point in $\mathbb{R}^2$ is represented by an ordered pair (x, y), where x and y denote its coordinates along the x- and y-axes, respectively. An alternative way to describe the location of a point is by using its distance from the origin and the angle it makes with the positive x-axis. This leads to the notion of *polar coordinates*.

Let $P = (x, y)$ be a point in $\mathbb{R}^2$, different from the origin. Let r denote the distance from the origin to P, and let θ denote the angle measured counterclockwise from the positive x-axis to the line segment joining the origin to P (see Figure 9). Then the pair (r, θ) is called the *polar coordinates* of P.

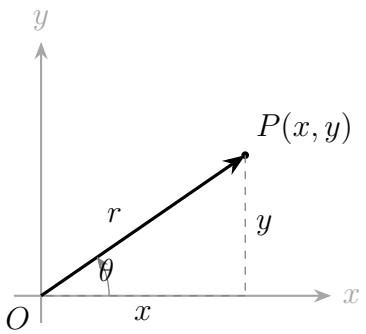


Figure 9: Polar coordinates in $\mathbb{R}^2$.

The Cartesian and polar coordinates are related by

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Conversely,

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right),$$

where the value of θ is chosen according to the quadrant in which the point (x, y) lies.

Throughout these notes, we shall assume

$$r \geq 0, \qquad 0 \leq \theta < 2\pi.$$

Observe that the polar coordinates of the origin are not unique, since $r = 0$ while θ may be chosen arbitrarily.

Cartesian coordinates. We now return to three-dimensional Euclidean space. A point in $\mathbb{R}^3$ is represented by its Cartesian coordinates (x, y, z), which specify its position relative to the three coordinate axes. Geometrically, these coordinates may be obtained by starting at the origin, moving a distance x along the x-axis, then a distance y parallel to the y-axis, and finally a distance z parallel to the z-axis (see Figure 10).

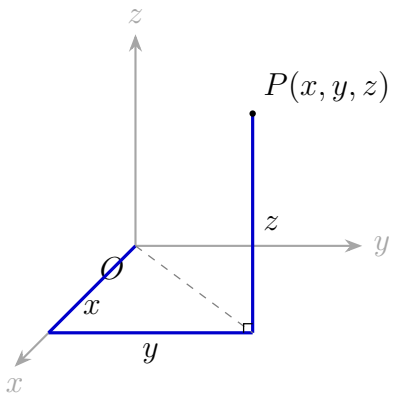


Figure 10: Cartesian coordinates in $\mathbb{R}^3$.

Although Cartesian coordinates are convenient for many problems, they are not always the most suitable choice. In many applications, the geometry of the problem possesses cylindrical or spherical symmetry. In such situations, it is often advantageous to describe a point using *curvilinear coordinates*, where one or more of the coordinates are determined by distances and angles rather than distances measured parallel to the coordinate axes.

Cylindrical coordinates. Let $P = (x, y, z)$ be a point in $\mathbb{R}^3$, and let $P_0 = (x, y, 0)$ denote the orthogonal projection of P onto the xy-plane. Treating (x, y) as a point in $\mathbb{R}^2$, let (r, θ) be its polar coordinates (see Figure 11). The triple (r, θ, z) is called the *cylindrical coordinates* of the point P.

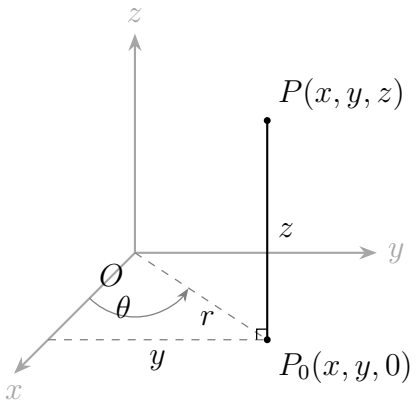


Figure 11: Cylindrical coordinates in $\mathbb{R}^3$.

The Cartesian and cylindrical coordinates are related by

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

Conversely,

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right), \qquad z = z,$$

where the value of θ is chosen according to the quadrant in which the point (x, y) lies.

Throughout these notes, we assume

$$r \geq 0, \qquad 0 \leq \theta < 2\pi.$$

Observe that θ is not defined when $(x, y) = (0, 0)$, that is, for points lying on the z-axis.

Spherical coordinates. Let $P = (x, y, z)$ be a point in $\mathbb{R}^3$, different from the origin. Let ρ denote the distance from the origin to P, and let θ be the angle that the projection of $\overrightarrow{OP}$ onto the xy-plane makes with the positive x-axis. Finally, let ϕ be the angle between the positive z-axis and the vector $\overrightarrow{OP}$ (see Figure 12). The triple (ρ, θ, ϕ) is called the *spherical coordinates* of the point P.

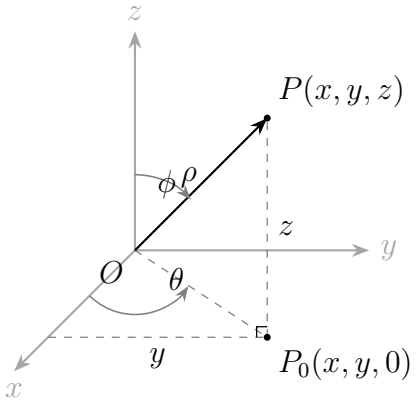


Figure 12: Spherical coordinates in $\mathbb{R}^3$.

The Cartesian and spherical coordinates are related by

$$x = \rho \sin\phi \cos\theta, \qquad y = \rho \sin\phi \sin\theta, \qquad z = \rho \cos\phi.$$

Conversely,

$$\rho = \sqrt{x^2 + y^2 + z^2}, \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right), \qquad \phi = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right),$$

where the value of θ is chosen according to the quadrant in which the point (x, y) lies.

Throughout these notes, we assume

$$\rho \geq 0, \qquad 0 \leq \theta < 2\pi, \qquad 0 \leq \phi \leq \pi.$$

Observe that θ is not defined for points on the z-axis, while the spherical coordinates of the origin are not uniquely determined since $\rho = 0$.

2 Multivariable calculus

We now extend our study of functions to several variables. In particular, we will consider functions whose inputs and outputs may be vectors and develop the basic ideas needed to study them.

2.1 Vector-valued and scalar functions

Definition 7. *A vector-valued function of a real variable is a rule that associates a vector* $\mathbf{f}(t)$ *with a real number* t*, where* t *is in some subset* D *of* $\mathbb{R}^1$ *(called the* ***domain of*** $\mathbf{f}$*). We write* $\mathbf{f} : D \to \mathbb{R}^3$ *to denote that* $\mathbf{f}$ *is a mapping of* D *into* $\mathbb{R}^3$*.*

A vector-valued function of a real variable can be written in component form as

$$\boxed{\mathbf{f}(t) = f_1(t)\mathbf{i} + f_2(t)\mathbf{j} + f_3(t)\mathbf{k}}$$

or equivalently,

$$\boxed{\mathbf{f}(t) = \big(f_1(t), f_2(t), f_3(t)\big)},$$

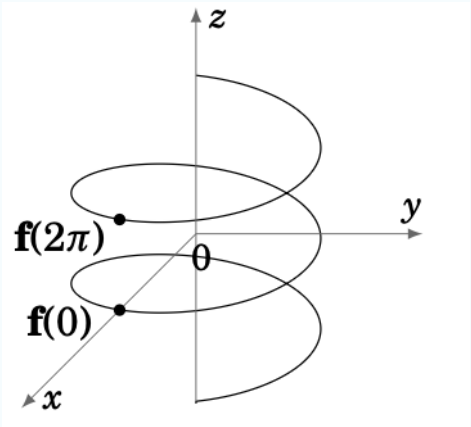
where $f_1(t)$, $f_2(t)$, and $f_3(t)$ are real-valued functions called the *component functions* of $\mathbf{f}$.

Example 1. *Define* $\mathbf{f} : \mathbb{R} \to \mathbb{R}^3$ *by*

$$\mathbf{f}(t) = (\cos t, \sin t, t).$$

This is the equation of a helix. *In particular,*

$$\mathbf{f}(0) = (1, 0, 0), \qquad \mathbf{f}(2\pi) = (1, 0, 2\pi).$$



Definition 8. *Let* $\mathbf{f}(t)$ *be a vector-valued function, let* a *be a real number, and let* $\mathbf{c}$ *be a vector. We say that*

$$\lim_{t\to a} \mathbf{f}(t) = \mathbf{c}$$

if

$$\lim_{t\to a} \|\mathbf{f}(t) - \mathbf{c}\| = 0.$$

If $\mathbf{f}(t) = (f_1(t), f_2(t), f_3(t))$, *then*

$$\lim_{t\to a} \mathbf{f}(t) = \left(\lim_{t\to a} f_1(t), \lim_{t\to a} f_2(t), \lim_{t\to a} f_3(t)\right),$$

provided that all three limits on the right-hand side exist.

Definition 9. *Let* $\mathbf{f}(t) = (f_1(t), f_2(t), f_3(t))$ *be a vector-valued function, and let* a *be a real number in its domain. Then* $\mathbf{f}(t)$ *is continuous at* a *if*

$$\lim_{t\to a} \mathbf{f}(t) = \mathbf{f}(a).$$

Equivalently, $\mathbf{f}(t)$ *is continuous at* a *if and only if* $f_1(t)$, $f_2(t)$, *and* $f_3(t)$ *are continuous at* a.

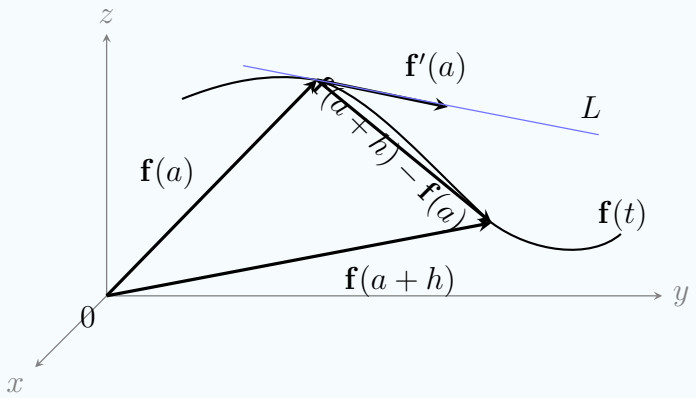
The derivative of $\mathbf{f}(t)$ *at* a, *denoted by* $\mathbf{f}'(a)$ *or* $\dfrac{d\mathbf{f}}{dt}(a)$, *is the limit*

$$\mathbf{f}'(a) = \lim_{h\to 0} \frac{\mathbf{f}(a+h) - \mathbf{f}(a)}{h},$$

if that limit exists. Equivalently,

$$\mathbf{f}'(a) = \big(f_1'(a), f_2'(a), f_3'(a)\big),$$

if the component derivatives exist. We say that $\mathbf{f}(t)$ *is differentiable at* a *if* $\mathbf{f}'(a)$ *exists.*



Recall that for a real-valued function of a single variable, the derivative represents the slope of the tangent line to the graph at a given point. For a vector-valued function, the derivative is a *tangent vector* to the curve represented by the function, and this vector lies along the *tangent line* to the curve at the corresponding point.

Example 2. *Let* $\mathbf{f}(t) = (\cos t, \sin t, t)$*. Then*

$$\mathbf{f}'(t) = (-\sin t, \cos t, 1)$$

for all t*. The tangent line* L *to the curve at* $\mathbf{f}(2\pi) = (1, 0, 2\pi)$ *is*

$$L = \mathbf{f}(2\pi) + s\mathbf{f}'(2\pi) = (1, 0, 2\pi) + s(0, 1, 1),$$

or, in parametric form,

$$x = 1, \qquad y = s, \qquad z = 2\pi + s, \qquad -\infty < s < \infty.$$

A *scalar function* is a real-valued function. If $u(t)$ is a scalar function and $\mathbf{f}(t)$ is a vector-valued function, then their product, defined by $(u\mathbf{f})(t) = u(t)\mathbf{f}(t)$, is a vector-valued function. The usual rules of differentiation and the basic properties of derivatives extend naturally to vector-valued functions.

Exercise. *Suppose* $\mathbf{f}(t)$ *is differentiable and* $\|\mathbf{f}(t)\| \neq 0$ *for all* t*. If* $\|\mathbf{f}(t)\|$ *is constant, show that*

$$\mathbf{f}(t) \cdot \mathbf{f}'(t) = 0 \qquad \text{for all } t.$$

2.2 Arc length

Arc length is useful in studying curvature and the motion of curves in differential geometry.

Let $\mathbf{r}(t) = (x(t), y(t), z(t))$ be the position vector of an object moving in $\mathbb{R}^3$. Its velocity is $\mathbf{r}'(t)$, so its speed at time t is $\|\mathbf{r}'(t)\|$. Therefore, the distance traveled by the object from $t = a$ to $t = b$ is

$$s = \int_a^b \|\mathbf{r}'(t)\|\, dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\, dt.$$

This also gives the **arc length** of the curve traced by $\mathbf{r}(t)$ between $t = a$ and $t = b$.

Definition 10. *Let* C *be a smooth curve in* $\mathbb{R}^3$ *represented by a function* $\mathbf{f}(t)$ *defined on an interval* $[a, b]$*, and let* $\alpha : [c, d] \to [a, b]$ *be a smooth one-to-one mapping of an interval* $[c, d]$ *onto* $[a, b]$*. Then the function* $\mathbf{g} : [c, d] \to \mathbb{R}^3$ *defined by* $\mathbf{g}(s) = \mathbf{f}(\alpha(s))$ *is a* **parametrization** *of* C *with* parameter s*. If* α *is strictly increasing on* $[c, d]$*, then we say that* $\mathbf{g}(s)$ *is* **equivalent** *to* $\mathbf{f}(t)$*. A parametrization* $\mathbf{f} : [a, b] \to \mathbb{R}^3$ *is called* **smooth** *if* $\mathbf{f}'(t)$ *exists and* $\mathbf{f}'(t) \neq \mathbf{0}$ *for all* $t \in [a, b]$*.*

Theorem 4 (Chain Rule)**.** *If* $\mathbf{f}(t)$ *is a differentiable vector-valued function of* t*, and* $t = \alpha(s)$ *is a differentiable scalar function of* s*, then* $\mathbf{f}(s) = \mathbf{f}(\alpha(s))$ *is a differentiable vector-valued function of* s*, and*

$$\frac{d\mathbf{f}}{ds} = \frac{d\mathbf{f}}{dt}\frac{dt}{ds}$$

for any s *where the composite function* $\mathbf{f}(\alpha(s))$ *is defined.*

Example 3. *Let* $\mathbf{r}(t) = (\cos t, \sin t, t)$*,* $0 \leq t \leq 2\pi$*, and consider the reparametrization* $t = 2s$*. Then*

$$\mathbf{g}(s) = \mathbf{r}(2s) = (\cos 2s, \sin 2s, 2s), \qquad 0 \leq s \leq \pi.$$

By the chain rule,

$$\mathbf{g}'(s) = \mathbf{r}'(2s)\frac{dt}{ds} = 2(-\sin 2s, \cos 2s, 1).$$

Hence, the arc length of the curve using the parameter s is

$$L = \int_0^{\pi} \|\mathbf{g}'(s)\|\, ds = \int_0^{\pi} 2\sqrt{2}\, ds = 2\pi\sqrt{2}.$$

The same curve using the original parameter gives

$$L = \int_0^{2\pi} \|\mathbf{r}'(t)\|\, dt = \int_0^{2\pi} \sqrt{2}\, dt = 2\pi\sqrt{2}.$$

Thus, equivalent parametrization gives the same arc length.

Example 4. *Consider the curve $\mathbf{r}(t) = (t, t^2, 0)$, $0 \le t \le 1$, and re-parametrize it by $t = s^3$, $0 \le s \le 1$. Then*

$$\mathbf{g}(s) = \mathbf{r}(s^3) = (s^3, s^6, 0).$$

Since $t = s^3$ is strictly increasing, $\mathbf{g}(s)$ is equivalent to $\mathbf{r}(t)$. By the chain rule,

$$\mathbf{g}'(s) = \mathbf{r}'(s^3)\frac{dt}{ds} = (1, 2s^3, 0)(3s^2) = (3s^2, 6s^5, 0).$$

Therefore, the arc length computed using the new parameter is

$$L = \int_0^1 \|\mathbf{g}'(s)\|\, ds = \int_0^1 3s^2\sqrt{1 + 4s^6}\, ds.$$

On the other hand, using the original parameter,

$$L = \int_0^1 \|\mathbf{r}'(t)\|\, dt = \int_0^1 \sqrt{1 + 4t^2}\, dt.$$

With the substitution $t = s^3$, these two expressions are equal, showing again that the arc length is independent of the choice of equivalent parametrization.

A curve may have different parametrizations, and the speed may depend on the choice of parameter. It is therefore useful to use the *arc length* as a parameter. Let $\mathbf{f}(t)$ be a smooth parametrization of a curve on $[a, b]$. Define

$$s = s(t) = \int_a^t \|\mathbf{f}'(u)\|\, du.$$

Here s represents the distance traveled along the curve from $t = a$ to t. In particular,

$$s(a) = 0, \qquad s(b) = L,$$

where L is the total arc length of the curve. By the Fundamental Theorem of Calculus,

$$s'(t) = \|\mathbf{f}'(t)\| > 0.$$

Hence $s(t)$ is strictly increasing and has a differentiable inverse $t = \alpha(s)$, where

$$\alpha'(s) = \frac{1}{\|\mathbf{f}'(\alpha(s))\|}.$$

We can therefore reparametrize the curve using the arc length parameter s by defining

$$\mathbf{f}(s) = \mathbf{f}(\alpha(s)), \qquad 0 \le s \le L.$$

By the Chain Rule,

$$\mathbf{f}'(s) = \mathbf{f}'(\alpha(s))\alpha'(s) = \frac{\mathbf{f}'(\alpha(s))}{\|\mathbf{f}'(\alpha(s))\|}.$$

Consequently,

$$\|\mathbf{f}'(s)\| = 1.$$

Thus, an **arc length parametrization** traverses the curve with ***unit speed***.

Theorem 5 (Inverse Function Theorem)**.** *Let $s : [a, b] \to \mathbb{R}$ be differentiable and one-to-one, with $s'(t) \neq 0$ for all $t \in [a, b]$. Then s has an inverse function*

$$t = \alpha(s),$$

defined on $s([a, b])$. The inverse function α is differentiable, and

$$\alpha'(s) = \frac{1}{s'(\alpha(s))}.$$

Example 5. *Parametrize the curve*

$$\mathbf{f}(t) = (\cos t, \sin t, 2t), \qquad t \in [0, 2\pi],$$

by arc length.
Solution: *We have*

$$\mathbf{f}'(t) = (-\sin t, \cos t, 2),$$

and hence

$$\|\mathbf{f}'(t)\| = \sqrt{\sin^2 t + \cos^2 t + 4} = \sqrt{5}.$$

Therefore, by the definition of arc length parameter,

$$s = \int_0^t \|\mathbf{f}'(u)\|\, du = \int_0^t \sqrt{5}\, du = \sqrt{5}\, t$$

for all $t \in [0, 2\pi]$.

Thus,

$$t = \alpha(s) = \frac{s}{\sqrt{5}}.$$

Since $t \in [0, 2\pi]$, the corresponding range of s is

$$0 \le s \le 2\pi\sqrt{5}.$$

Hence the required arc length parametrization is

$$\mathbf{f}(s) = \left(\cos\frac{s}{\sqrt{5}}, \sin\frac{s}{\sqrt{5}}, \frac{2s}{\sqrt{5}}\right), \qquad s \in [0, 2\pi\sqrt{5}].$$

Indeed,

$$\|\mathbf{f}'(s)\| = 1.$$

For a smooth curve $\mathbf{f}(t)$ with $\mathbf{f}'(t) \neq \mathbf{0}$, the vector $\mathbf{f}'(t)$ gives the **tangent direction** to the curve. The corresponding **unit tangent vector** is

$$\mathbf{T}(t) = \frac{\mathbf{f}'(t)}{\|\mathbf{f}'(t)\|}.$$

The direction in which the tangent vector changes is described by the **principal normal vector**. Its unit version is given by

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} = \frac{\mathbf{f}'(t) \times \big(\mathbf{f}''(t) \times \mathbf{f}'(t)\big)}{\|\mathbf{f}'(t)\| \, \|\mathbf{f}''(t) \times \mathbf{f}'(t)\|}.$$

The **unit binormal vector** is defined by

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t) = \frac{\mathbf{f}'(t) \times \mathbf{f}''(t)}{\|\mathbf{f}'(t) \times \mathbf{f}''(t)\|}.$$

Thus, $\mathbf{T}(t)$, $\mathbf{N}(t)$, and $\mathbf{B}(t)$ form a right-handed system of mutually perpendicular unit vectors, called an **orthonormal frame**, attached to the curve at each point. Finally, the **curvature** $\kappa(t)$ measures how much the curve bends at a given point and is given by

$$\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{f}'(t)\|} = \frac{\|\mathbf{f}'(t) \times \mathbf{f}''(t)\|}{\|\mathbf{f}'(t)\|^3}.$$

Thus, a larger value of $\kappa(t)$ indicates a sharper bend, while a smaller value indicates that the curve is flatter. Check these quantities explicitly for the helix $\mathbf{f}(t) = (\cos t, \sin t, t)$!

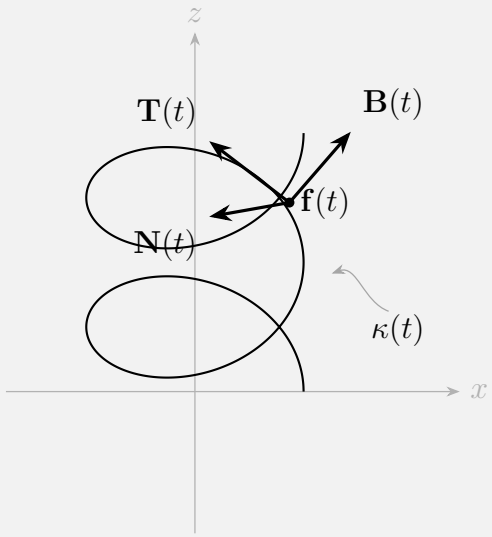


Figure 13: Frenet frame along a curve

Lecture 3, October 12, 2026

2.3 Functions of two or three variables

A real-valued function f defined on a subset D of $\mathbb{R}^2$ is a rule that assigns to each point (x, y) in D a real number $f(x, y)$. The largest possible set D in $\mathbb{R}^2$ on which f is defined is called the **domain** of f, and the **range** of f is the set of all real numbers $f(x, y)$ as (x, y) varies over the domain D. A similar definition holds for functions $f(x, y, z)$ defined on points (x, y, z) in $\mathbb{R}^3$.

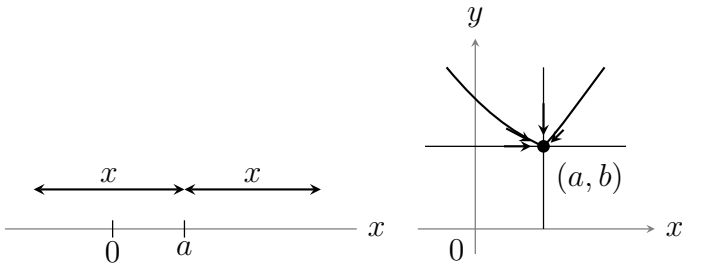
Example 6. *Consider the following functions and determine their domains and ranges.*

1. *Let* $f(x, y) = \dfrac{1}{x - y}$. *The domain is* $D = \{(x, y) \in \mathbb{R}^2 : x \neq y\}$, *and the range is* $\mathbb{R} \setminus \{0\}$.
2. *Let* $f(x, y) = \sqrt{1 - x^2 - y^2}$. *The domain is* $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$, *and the range is* $[0, 1]$.
3. *Let* $f(x, y, z) = e^{x+y-z}$. *The domain is* $D = \mathbb{R}^3$, *and the range is* $(0, \infty)$.

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function of two variables. We usually write $z = f(x, y)$. The **graph** of f is the set of points $G_f = \{(x, y, z) \in \mathbb{R}^3 : z = f(x, y)\}$. Thus, the graph of a function of two variables is a surface in $\mathbb{R}^3$. Equivalently, if we define $F(x, y, z) =$

$f(x,y) - z$, then the graph can be written as $G_f = \{(x,y,z) \in \mathbb{R}^3 : F(x,y,z) = 0\}$.

In one variable, when we write $x \to a$, the point x can approach a only from the two directions along the real line, as illustrated in (a) below. In two variables, however, a point (x,y) can approach (a,b) from infinitely many directions and along infinitely many paths, as shown in (b) below. Therefore, the idea of a limit in several variables must account for all possible ways of approaching the point. To capture this geometrically, instead of considering an interval around a, we consider a disk centered at (a,b). If (x,y) is sufficiently close to (a,b), it lies inside this disk regardless of the direction or path along which it approaches (a,b). This motivates the definition of the limit of a function of several variables: the value of the function should approach the same number as (x,y) approaches (a,b) from every possible direction and along every possible path.



(a) $x \to a$ in $\mathbb{R}$ **(b)** $(x,y) \to (a,b)$ in $\mathbb{R}^2$

Definition 11. *Let $(a,b) \in \mathbb{R}^2$ and let $\delta > 0$. The **deleted neighborhood** of (a,b) with radius δ is the set of all points $(x,y) \in \mathbb{R}^2$ satisfying*

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta.$$

Thus, it consists of all points inside the disk of radius δ centered at (a,b), except the point (a,b) itself.

Definition 12. *Let $(a,b) \in \mathbb{R}^2$, and let f be a real-valued function defined at all points (x,y) in some deleted neighborhood of (a,b). We say that the limit of $f(x,y)$ as (x,y) approaches (a,b) is L, written as*

$$\lim_{(x,y)\to(a,b)} f(x,y) = L,$$

if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$|f(x,y) - L| < \epsilon$$

whenever

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta.$$

In many simple cases, the limit of a function of several variables can be found by direct substitution. If $f(x, y)$ is defined at (a, b) and its formula does not lead to an indeterminate form $0/0$ at that point, then

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b).$$

Example 7. *Consider*

$$\lim_{(x,y)\to(1,2)} \frac{xy}{x^2 + y^2}.$$

Since the function is defined at $(1, 2)$*, we can substitute* $x = 1$ *and* $y = 2$ *directly:*

$$\lim_{(x,y)\to(1,2)} \frac{xy}{x^2 + y^2} = \frac{(1)(2)}{1^2 + 2^2} = \frac{2}{5}.$$

Example 8. *Consider*

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2}.$$

Direct substitution gives the indeterminate form $0/0$*. To check whether the limit exists, consider different paths approaching* $(0, 0)$*.*
Along the x*-axis,* $y = 0$*, so*

$$\frac{xy}{x^2 + y^2} = 0.$$

Along the line $y = x$*, we obtain*

$$\frac{xy}{x^2 + y^2} = \frac{x^2}{2x^2} = \frac{1}{2}.$$

Since the function approaches different values along these two paths, the limit does not exist.

Theorem 6. *Suppose that* $\lim_{(x,y)\to(a,b)} f(x, y)$ *and* $\lim_{(x,y)\to(a,b)} g(x, y)$ *both exist, and that* k *is some scalar. Then:*

(a)

$$\lim_{(x,y)\to(a,b)} [f(x, y) \pm g(x, y)] = \left[\lim_{(x,y)\to(a,b)} f(x, y)\right] \pm \left[\lim_{(x,y)\to(a,b)} g(x, y)\right].$$

(b)

$$\lim_{(x,y)\to(a,b)} kf(x, y) = k\left[\lim_{(x,y)\to(a,b)} f(x, y)\right].$$

(c)

$$\lim_{(x,y)\to(a,b)}[f(x,y)g(x,y)] = \left[\lim_{(x,y)\to(a,b)} f(x,y)\right]\left[\lim_{(x,y)\to(a,b)} g(x,y)\right].$$

(d)

$$\lim_{(x,y)\to(a,b)} \frac{f(x,y)}{g(x,y)} = \frac{\lim_{(x,y)\to(a,b)} f(x,y)}{\lim_{(x,y)\to(a,b)} g(x,y)} \qquad \text{if} \qquad \lim_{(x,y)\to(a,b)} g(x,y) \neq 0.$$

(e) If $|f(x,y) - L| \leq g(x,y)$ *for all* (x,y) *and if*

$$\lim_{(x,y)\to(a,b)} g(x,y) = 0,$$

then

$$\lim_{(x,y)\to(a,b)} f(x,y) = L.$$

Example 9. *Show that*

$$\lim_{(x,y)\to(0,0)} \frac{x^4}{x^2+y^2} = 0.$$

Since direct substitution gives the indeterminate form $0/0$*, we use Theorem 6(e). For all* $(x,y) \neq (0,0)$*,*

$$0 \leq \left|\frac{x^4}{x^2+y^2}\right| \leq \frac{(x^2+y^2)^2}{x^2+y^2} = x^2+y^2 \to 0$$

as $(x,y) \to (0,0)$*, where we used* $x^4 \leq (\sqrt{x^2})^4 \leq (\sqrt{x^2+y^2})^4$ *for all* (x,y)*. Therefore,*

$$\lim_{(x,y)\to(0,0)} \frac{x^4}{x^2+y^2} = 0.$$

Definition 13. *A real-valued function* $f(x,y)$ *with domain* D *in* $\mathbb{R}^2$ *is* ***continuous*** *at the point* (a,b) *in* D *if*

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b).$$

We say that $f(x,y)$ *is a* ***continuous function*** *if it is continuous at every point in its domain* D*.*

Example 10. *Define* $f : \mathbb{R}^2 \to \mathbb{R}$ *by*

$$f(x,y) = \begin{cases} \dfrac{x^2 y}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

For $(x,y) \neq (0,0)$*, we have*

$$\left| \frac{x^2 y}{x^2+y^2} \right| \leq |y| \leq \sqrt{x^2+y^2} \to 0$$

as $(x,y) \to (0,0)$*. Hence*

$$\lim_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0).$$

Therefore, f *is continuous at* $(0,0)$*. At every other point,* f *is continuous by direct substitution. Thus,* f *is continuous on* $\mathbb{R}^2$*.*

Example 11. *Consider*

$$\lim_{(x,y)\to(0,0)} \frac{x}{y}.$$

Direct substitution gives the indeterminate form $0/0$*. To check the limit, consider different paths approaching* $(0,0)$*. Along the line* $y = x$*,*

$$\frac{x}{y} = 1,$$

whereas along the line $y = 2x$*,*

$$\frac{x}{y} = \frac{1}{2}.$$

Since the function approaches different values along different paths, the limit does not exist.

Example 12. *Consider*

$$\lim_{(x,y)\to(0,0)} \frac{y^4 \sin(xy)}{x^2+y^2}.$$

Since $|\sin(xy)| \leq 1$*,*

$$\left| \frac{y^4 \sin(xy)}{x^2+y^2} \right| \leq \frac{y^4}{x^2+y^2} \leq \frac{y^2(y^2+x^2)}{x^2+y^2} \leq y^2 \to 0$$

as $(x,y) \to (0,0)$*. Therefore,*

$$\lim_{(x,y)\to(0,0)} \frac{y^4 \sin(xy)}{x^2+y^2} = 0.$$

Example 13. *Use the substitution* $r = \sqrt{x^2 + y^2}$ *to show that*

$$\lim_{(x,y)\to(0,0)} \frac{\sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}} = 1.$$

Solution: *Let* $r = \sqrt{x^2 + y^2}$. *Then* $(x, y) \to (0, 0)$ *implies* $r \to 0$, *and the limit becomes*

$$\lim_{r\to 0} \frac{\sin r}{r} = 1.$$

This is the standard single-variable limit, which can be obtained, for example, using L'Hôpital's Rule:

$$\lim_{r\to 0} \frac{\sin r}{r} = \lim_{r\to 0} \frac{\cos r}{1} = 1.$$

Hence,

$$\lim_{(x,y)\to(0,0)} \frac{\sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}} = 1.$$

2.4 Partial derivatives

When a function depends on several variables, we may want to study how it changes with respect to one variable while keeping the other variables fixed. This leads to the idea of a **partial derivative**. For a function $f(x, y)$, we can therefore differentiate with respect to x while keeping y fixed, or with respect to y while keeping x fixed.

Definition 14. *Let* $f(x, y)$ *be a real-valued function with domain* D *in* $\mathbb{R}^2$, *and let* (a, b) *be a point in* D. *Then the* ***partial derivative*** *of* f *at* (a, b) *with respect to* x, *denoted by* $\dfrac{\partial f}{\partial x}(a, b)$, *is defined as*

$$\frac{\partial f}{\partial x}(a,b) = \lim_{h\to 0} \frac{f(a+h,b) - f(a,b)}{h}.$$

The ***partial derivative*** *of* f *at* (a, b) *with respect to* y, *denoted by* $\dfrac{\partial f}{\partial y}(a, b)$, *is defined as*

$$\frac{\partial f}{\partial y}(a,b) = \lim_{h\to 0} \frac{f(a,b+h) - f(a,b)}{h}.$$

Since $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ are themselves functions of x and y, we can differentiate them again.

This gives the **higher-order partial derivatives**. For example,

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = f_{xx} = D_{11}f, \qquad \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = f_{yy} = D_{22}f,$$

and the mixed partial derivatives are

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = f_{xy} = D_{12}f, \qquad \frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = f_{yx} = D_{21}f.$$

Higher-order partial derivatives are obtained similarly by repeatedly differentiating.

Example 14. *Let*

$$f(x, y) = x^2 - y^2 + 6xy + 4x - 8y + 2.$$

The first-order partial derivatives are

$$\frac{\partial f}{\partial x} = 2x + 6y + 4, \qquad \frac{\partial f}{\partial y} = -2y + 6x - 8.$$

The second-order partial derivatives are

$$\frac{\partial^2 f}{\partial x^2} = 2, \qquad \frac{\partial^2 f}{\partial y^2} = -2,$$

and

$$\frac{\partial^2 f}{\partial y\,\partial x} = 6, \qquad \frac{\partial^2 f}{\partial x\,\partial y} = 6.$$

Since the second-order mixed partial derivatives are equal, we have

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y} = 6.$$

The third-order partial derivatives are all zero:

$$\frac{\partial^3 f}{\partial x^3} = \frac{\partial^3 f}{\partial y^3} = \frac{\partial^3 f}{\partial y\,\partial x^2} = \frac{\partial^3 f}{\partial x\,\partial y^2} = \frac{\partial^3 f}{\partial y^2\,\partial x} = \frac{\partial^3 f}{\partial x^2\,\partial y} = \frac{\partial^3 f}{\partial x\,\partial y\,\partial x} = \frac{\partial^3 f}{\partial y\,\partial x\,\partial y} = 0.$$

In general, the mixed partial derivatives f_{xy} and f_{yx} need not be equal. However, if the second-order partial derivatives f_{xy} and f_{yx} are continuous in a neighborhood of a point, then

$$f_{xy} = f_{yx}$$

at that point. This result is known as **Clairaut's theorem** (or **Schwarz's theorem**). The equality need not hold when the required continuity condition is absent. For

example, define

$$f(x,y) = \begin{cases} \dfrac{xy(x^2-y^2)}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Then, at $(0,0)$,

$$f_{xy}(0,0) = -1, \qquad f_{yx}(0,0) = 1.$$

Thus,

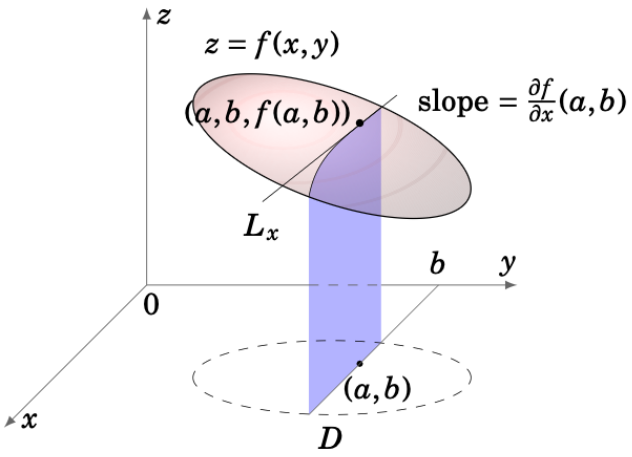
$$f_{xy}(0,0) \neq f_{yx}(0,0).$$

2.5 Tangent plane to a surface

Recall that the derivative $\dfrac{dy}{dx}$ of a function $y = f(x)$ has a geometric meaning, namely, as the slope of the tangent line to the graph of f at the point $(x, f(x))$ in $\mathbb{R}^2$. In particular, at $x = a$, the slope of the tangent line is $f'(a)$. Since the tangent line passes through $(a, f(a))$, its equation is

$$y - f(a) = f'(a)(x - a).$$

We now seek a similar geometric interpretation for the partial derivatives of a function of two variables. The partial derivatives $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ can be thought of as the rate of change of a function $z = f(x,y)$ in the positive x and y directions, respectively. Let $(a, b, f(a,b))$ be a point on the surface $z = f(x,y)$. Then we want to find the equation of the tangent plane at that point. First, look at the image (a).



(a) Tangent line L_x in the plane $y = b$

The equation of the tangent line L_x will be as follows: This line stays in the plane $y = b$,

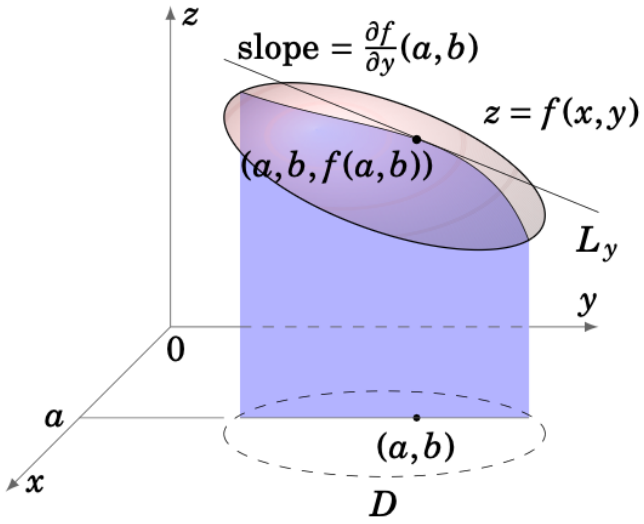
and so x and z vary. Therefore, the equation will be

$$z - f(a,b) = \frac{\partial f}{\partial x}(a,b)(x-a), \qquad y = b.$$

Hence, in the vector equation form, it can be written as

$$(x,y,z) = (a,b,f(a,b)) + t(1,0,\frac{\partial f}{\partial x}(a,b)).$$

Now, look at the second image (b).

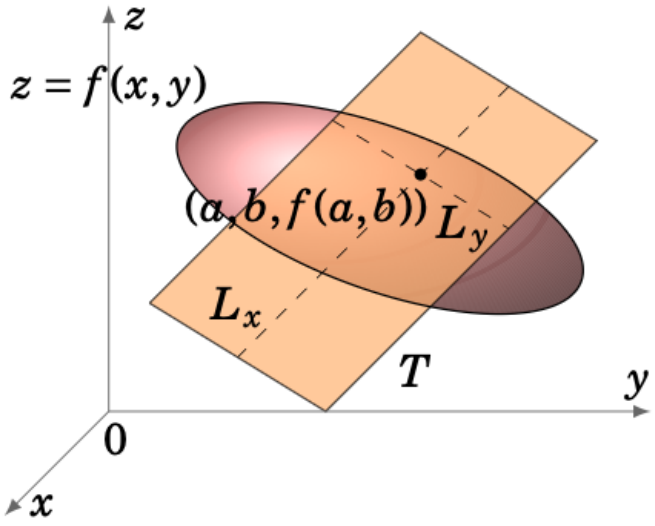


(b) Tangent line L_y in the plane $x = a$

So, in the vector equation form, the equation of L_y can be written as

$$(x,y,z) = (a,b,f(a,b)) + t(0,1,\frac{\partial f}{\partial y}(a,b)).$$

Now the tangent plane is



This contains the lines L_x and L_y. Then, it contains the vectors

$$v_x = (1, 0, \frac{\partial f}{\partial x}(a, b)) \qquad \text{and} \qquad v_y = (0, 1, \frac{\partial f}{\partial y}(a, b)),$$

and so it is perpendicular to the normal vector

$$\mathbf{n} = v_x \times v_y = \left(-\frac{\partial f}{\partial x}(a, b), -\frac{\partial f}{\partial y}(a, b), 1 \right).$$

Hence, in terms of the vector equation, the tangent plane will be

$$\mathbf{n} \cdot ((x, y, z) - (a, b, f(a, b))) = 0.$$

So, the equation of the tangent plane to the surface $z = f(x, y)$ at the point $(a, b, f(a, b))$ is

$$\frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b) - z + f(a, b) = 0.$$

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Suppose a surface is defined implicitly by

$$F(x, y, z) = 0$$

and passes through (a, b, c). The gradient

$$\nabla F(a, b, c) = \big(F_x(a, b, c), F_y(a, b, c), F_z(a, b, c)\big)$$

is normal to the surface at (a, b, c). Hence, the tangent plane is

$$\boxed{\nabla F(a, b, c) \cdot (x - a, y - b, z - c) = 0,}$$

or equivalently,

$$\boxed{F_x(a, b, c)(x - a) + F_y(a, b, c)(y - b) + F_z(a, b, c)(z - c) = 0.}$$

Example 15. *Find the equation of the tangent plane to the surface $z = xe^y$ at the point $(1, 0, 1)$.*

Solution: *For $f(x, y) = xe^y$, we have $\dfrac{\partial f}{\partial x} = e^y$ and $\dfrac{\partial f}{\partial y} = xe^y$. Thus, at $(1, 0)$,*

$f_x(1,0) = 1$ *and* $f_y(1,0) = 1$. *Therefore, the tangent plane is*

$$(x-1) + (y-0) - z + 1 = 0,$$

or

$$x + y - z = 0.$$

Example 16. *Find the equation of the tangent plane to the surface* $x^2 + y^2 - z^2 = 0$ *at the point* $P = (3,4,5)$.
Solution: *Let* $F(x,y,z) = x^2 + y^2 - z^2$. *Then* $F_x = 2x$, $F_y = 2y$, *and* $F_z = -2z$. *At* $P = (3,4,5)$, *we have* $F_x = 6$, $F_y = 8$, *and* $F_z = -10$. *Therefore, the tangent plane is*

$$6(x-3) + 8(y-4) - 10(z-5) = 0,$$

or

$$3x + 4y - 5z = 0.$$

2.6 Directional derivatives and the gradient

For a function $z = f(x,y)$, the partial derivatives $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ describe the instantaneous rate of change of f in the positive x- and y-directions, respectively. But what if we want to measure the rate of change in an arbitrary direction? This leads to the idea of the **directional derivative**, which gives the rate of change of f in any specified direction. Thus, while partial derivatives describe changes along the coordinate directions, directional derivatives allow us to study changes along any direction in the plane.

Definition 15. *Let* $f(x,y)$ *be a real-valued function with domain* D *in* $\mathbb{R}^2$, *and let* (a,b) *be a point in* D. *Let* $\mathbf{v}$ *be a unit vector in* $\mathbb{R}^2$. *The* ***directional derivative*** *of* f *at* (a,b) *in the direction of* $\mathbf{v}$, *denoted by* $D_{\mathbf{v}}f(a,b)$, *is defined as*

$$D_{\mathbf{v}}f(a,b) = \lim_{h \to 0} \frac{f((a,b) + h\mathbf{v}) - f(a,b)}{h}.$$

If $f(x,y)$ has continuous partial derivatives $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ in D, then the directional derivative has a simple formula. Let us first recall the **Mean Value Theorem**: If g is continuous on $[a,b]$ and differentiable on (a,b), then there exists $\theta \in (0,1)$ such that

$$g(b) - g(a) = g'\big(a + \theta(b-a)\big)(b-a).$$

Theorem 7. *Let $f(x, y)$ be a real-valued function with domain D in $\mathbb{R}^2$ such that the partial derivatives $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ exist and are continuous in D. Let (a, b) be a point in D, and let $\mathbf{v} = (v_1, v_2)$ be a unit vector in $\mathbb{R}^2$. Then*

$$D_{\mathbf{v}} f(a, b) = v_1 \frac{\partial f}{\partial x}(a, b) + v_2 \frac{\partial f}{\partial y}(a, b).$$

Proof. Let $\mathbf{v} = (v_1, v_2)$ and fix $h \neq 0$. We split the change in f into two parts:

$$f(a+hv_1, b+hv_2) - f(a, b) = \big[f(a+hv_1, b+hv_2) - f(a+hv_1, b)\big] + \big[f(a+hv_1, b) - f(a, b)\big].$$

For the first part, keep $x = a + hv_1$ fixed and apply the Mean Value Theorem to the function of one variable obtained by varying y. Thus, for some number α between 0 and 1,

$$f(a + hv_1, b + hv_2) - f(a + hv_1, b) = hv_2 \frac{\partial f}{\partial y}(a + hv_1, b + \alpha hv_2).$$

Similarly, applying the Mean Value Theorem to the second part, there exists a number β between 0 and 1 such that

$$f(a + hv_1, b) - f(a, b) = hv_1 \frac{\partial f}{\partial x}(a + \beta hv_1, b).$$

Therefore,

$$\frac{f(a + hv_1, b + hv_2) - f(a, b)}{h} = v_2 \frac{\partial f}{\partial y}(a + hv_1, b + \alpha hv_2) + v_1 \frac{\partial f}{\partial x}(a + \beta hv_1, b).$$

As $h \to 0$, both points on the right approach (a, b). Since the partial derivatives are continuous,

$$D_{\mathbf{v}} f(a, b) = v_1 \frac{\partial f}{\partial x}(a, b) + v_2 \frac{\partial f}{\partial y}(a, b).$$

□

The directional derivative can be written in a particularly simple form using the **gradient**. For a real-valued function $f(x, y)$, the gradient is the vector

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right).$$

Similarly, for a function $f(x, y, z)$, the gradient is

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right).$$

The symbol ∇ is called the **del operator**. Since the gradient contains all the first-order partial derivatives, the directional derivative in the direction of a unit vector $\mathbf{v}$

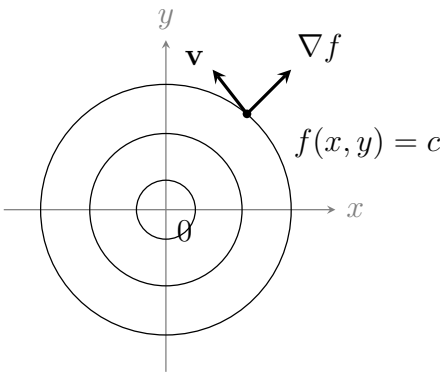
can be written as

$$D_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f.$$

Example 17. *Find the directional derivative of* $f(x, y) = x^2 + xy + y^2$ *at the point* $(1, -1)$ *in the direction of* $\mathbf{v} = (2, 1)$.
Solution: *Always normalize first. The unit vector in the given direction is* $\mathbf{u} = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$. *Since* $\nabla f = (2x + y, x + 2y)$, *we have* $\nabla f(1, -1) = (1, -1)$. *Therefore,*

$$D_{\mathbf{u}} f(1, -1) = \mathbf{u} \cdot \nabla f(1, -1) = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right) \cdot (1, -1) = \frac{1}{\sqrt{5}}.$$

A real-valued function $z = f(x, y)$ whose partial derivatives $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ exist and are continuous is called **continuously differentiable**. Assume that $f(x, y)$ is continuously differentiable and that $\nabla f \neq \mathbf{0}$. Let c be a real number in the range of f, and let $\mathbf{v}$ be a unit vector tangent to the level curve $f(x, y) = c$, as shown in the figure below.



Since $f(x, y)$ is constant along a level curve $f(x, y) = c$, its rate of change in any tangent direction to the curve is zero. But

$$D_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f = \|\mathbf{v}\| \, \|\nabla f\| \cos\theta,$$

where θ is the angle between $\mathbf{v}$ and ∇f. For a unit tangent vector $\mathbf{v}$, we have $\|\mathbf{v}\| = 1$, and since $\nabla f \neq \mathbf{0}$, the condition $D_{\mathbf{v}} f = 0$ gives $\cos\theta = 0$. Hence $\theta = 90°$, so ∇f is perpendicular to the level curve. Thus, the gradient gives the normal direction to a level curve.

Moreover, for any unit vector $\mathbf{v}$, we have $D_{\mathbf{v}} f = \|\nabla f\| \cos\theta$. At a fixed point, $\|\nabla f\|$ is fixed, so the directional derivative is largest when $\theta = 0°$, that is, in the direction of ∇f, and smallest when $\theta = 180°$, that is, in the direction of $-\nabla f$. Therefore, f increases most rapidly in the direction of ∇f and decreases most rapidly in the direction of $-\nabla f$.

Although the above discussion is for functions of two variables, the same ideas extend naturally to functions of three or more variables. In particular, for a function $f(x, y, z)$, the

directional derivative in the direction of a unit vector $\mathbf{v}$ is given by

$$D_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f.$$

The same interpretation of the gradient therefore remains valid: ∇f gives the direction of fastest increase, while $-\nabla f$ gives the direction of fastest decrease.

Theorem 8. *Let $f(x, y)$ be a continuously differentiable real-valued function, with $\nabla f \neq \mathbf{0}$. Then:*

(a) The gradient ∇f is normal to any level curve $f(x, y) = c$.

(b) The value of $f(x, y)$ increases fastest in the direction of ∇f.

(c) The value of $f(x, y)$ decreases fastest in the direction of $-\nabla f$.

Example 18. *The temperature at a point (x, y, z) in a room is given by $T(x, y, z) = x^2 + 2y^2 + z^2$. In which direction from the point $(1, -1, 2)$ will the temperature decrease most rapidly?*
Solution: *We have $\nabla T = (2x, 4y, 2z)$, so $\nabla T(1, -1, 2) = (2, -4, 4)$. Since the temperature decreases most rapidly in the direction of $-\nabla T$, the required direction is*

$$-\nabla T(1, -1, 2) = (-2, 4, -4).$$

The corresponding unit direction vector is

$$\frac{1}{6}(-2, 4, -4) = \left(-\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}\right).$$

2.7 Maxima and minima

The gradient can also be used to find **extreme points** of a real-valued function, that is, points where the function has a **local maximum** or **local minimum**. We will focus on functions of two variables. The functions of three or more variables require methods using linear algebra, such as matrices, quadratic forms, and eigenvalues and eigenvectors.

Since the gradient describes the direction of fastest increase of a function, it is also useful for locating points where the function does not increase or decrease in any direction. This leads naturally to the study of local maxima and minima.

Definition 16. *Let $f(x, y)$ be a real-valued function, and let (a, b) be a point in the domain of f. We say that f has a **local maximum** at (a, b) if there exists $r > 0$*

such that $f(x,y) \le f(a,b)$ *whenever* $(x-a)^2+(y-b)^2 < r^2$. *Similarly,* f *has a* ***local minimum*** *at* (a,b) *if there exists* $r > 0$ *such that* $f(x,y) \ge f(a,b)$ *whenever* $(x-a)^2+(y-b)^2 < r^2$.
If $f(x,y) \le f(a,b)$ *for every* (x,y) *in the domain of* f, *then* f *has a* ***global maximum*** *at* (a,b). *Similarly, if* $f(x,y) \ge f(a,b)$ *for every* (x,y) *in the domain of* f, *then* f *has a* ***global minimum*** *at* (a,b).

Suppose that (a,b) is a local maximum or minimum point of $f(x,y)$. By fixing $y = b$, we obtain a single-variable function $g(x) = f(x,b)$, which has a local maximum or minimum at $x = a$. Hence, $g'(a) = 0$, giving $\dfrac{\partial f}{\partial x}(a,b) = 0$. Similarly, by fixing $x = a$, we obtain $\dfrac{\partial f}{\partial y}(a,b) = 0$. Therefore,

$$\nabla f(a,b) = \mathbf{0}.$$

Theorem 9. *Let* $f(x,y)$ *be a real-valued function such that* $\dfrac{\partial f}{\partial x}(a,b)$ *and* $\dfrac{\partial f}{\partial y}(a,b)$ *exist. A necessary condition for* f *to have a local maximum or minimum at* (a,b) *is*

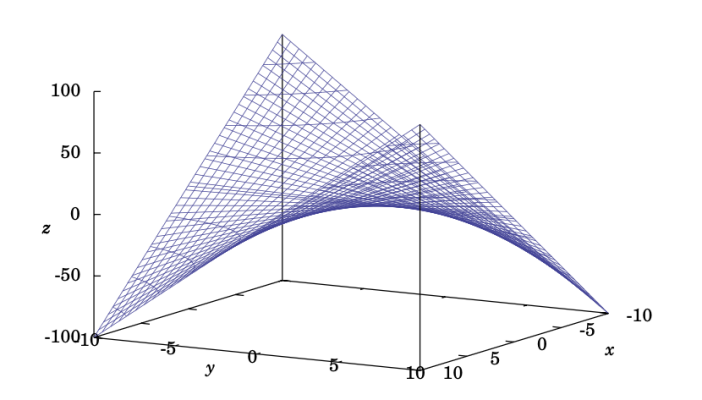
$$\nabla f(a,b) = \mathbf{0}.$$

A point (a,b) satisfying $\nabla f(a,b) = \mathbf{0}$ is called a **critical point** of f. Thus, to find the critical points, we solve

$$\frac{\partial f}{\partial x} = 0, \qquad \frac{\partial f}{\partial y} = 0$$

simultaneously. Note that $\nabla f(a,b) = \mathbf{0}$ is only a **necessary condition**; a critical point need not be a local maximum or minimum.

Example 19. *Consider the function* $f(x,y) = xy$. *Then* $\dfrac{\partial f}{\partial x} = y$ *and* $\dfrac{\partial f}{\partial y} = x$, *so the only critical point is* $(0,0)$. *However,* $(0,0)$ *is neither a local maximum nor a local minimum. Along the path* $y = x$, *we have* $f(x,y) = x^2 > 0 = f(0,0)$ *for* $x \neq 0$, *while along the path* $y = -x$, *we have* $f(x,y) = -x^2 < 0 = f(0,0)$. *Thus, the function takes both larger and smaller values arbitrarily close to* $(0,0)$.
Hence, $(0,0)$ *is a* ***saddle point****: the function increases in some directions and decreases in other directions. The graph of* $f(x,y)$ *is shown in the figure.*



Theorem 10. *Let $f(x,y)$ be a twice continuously differentiable real-valued function, with a critical point at (a,b), that is, $\nabla f(a,b) = \mathbf{0}$. Define*

$$D = \frac{\partial^2 f}{\partial x^2}(a,b)\frac{\partial^2 f}{\partial y^2}(a,b) - \left(\frac{\partial^2 f}{\partial y \partial x}(a,b)\right)^2.$$

Then

(a) if $D > 0$ and $\dfrac{\partial^2 f}{\partial x^2}(a,b) > 0$, then f has a local minimum at (a,b);

(b) if $D > 0$ and $\dfrac{\partial^2 f}{\partial x^2}(a,b) < 0$, then f has a local maximum at (a,b);

*(c) if $D < 0$, then f has a **saddle point** at (a,b);*

(d) if $D = 0$, then the test fails.

Example 20. *Find all local maxima and minima of*

$$f(x,y) = x^3 + y^3 - 3xy.$$

Solution: *We first find the critical points. Since $f_x = 3x^2 - 3y$ and $f_y = 3y^2 - 3x$, the critical points satisfy*

$$x^2 - y = 0, \qquad y^2 - x = 0.$$

Thus $y = x^2$ and $x = y^2$, giving the critical points $(0,0)$ and $(1,1)$.
The second-order partial derivatives are $f_{xx} = 6x$, $f_{yy} = 6y$, and $f_{yx} = -3$. Hence

$$D = f_{xx}f_{yy} - (f_{yx})^2 = 36xy - 9.$$

*At $(0,0)$, $D = -9 < 0$, so $(0,0)$ is a **saddle point**.*

At $(1,1)$*,* $D = 27 > 0$ *and* $f_{xx}(1,1) = 6 > 0$*. Therefore,* $(1,1)$ *is a* ***local minimum****. Hence,* f *has a local minimum at* $(1,1)$ *and a saddle point at* $(0,0)$*, and it has no local maximum.*